

PRISM: Provenance-Routed Inference via Specialist Models for Fake News Detection

1. Introduction

1.1 Problem: Detection That Presupposes Provenance Is Known

Fake news is not a monolithic category. Tandoc et al. (2018) reviewed 34 scholarly definitions of the term and organized them along two axes, facticity and the author's intent to deceive, yielding six recognizable types: news satire, news parody, fabrication, manipulation, native advertising, and propaganda. This paper is concerned with only one cell of that typology: fabrication, articles with no factual basis presented as legitimate news. Within fabrication, however, a further distinction has recently become consequential. Large language models (LLMs) can now produce fabricated news at a scale and fluency that rivals human writers, and the benchmarks built to study this shift, PolitiFact++ and GossipCop++, pair each human-written article with an LLM-generated counterpart, yielding four subclasses: human-written fake (HF), human-written real (HR), machine-generated fake (MF), and machine-generated real (MR) (Su et al., 2024). Tandoc et al.'s (2018) typology, formulated before this shift, has no axis for who produced a fabrication — only for how factual it is and how deceptive its intent. That silence sets up the question this paper investigates directly, using LIFE (Wang et al., 2026), a detector purpose-built for the LLM-generated case: does a system trained to separate fake from real actually learn a signal for veracity, or does it learn a signal for provenance, and treat the two as though they were the same thing?

1.2 Contributions

This paper makes four contributions. First, it reproduces LIFE's paper-faithful classifier head on the PolitiFact++ benchmark and confirms it approaches the paper's reported target. Second, it

runs a diagnostic chain — combining human and machine-written articles into a single training population, decomposing the resulting failure by class, and finally isolating human-only data entirely — that shows LIFE cannot detect human-written fake news at any point along that chain. Third, it builds a practical system, PRISM, around this finding: a provenance router that sends LLM-generated articles to LIFE and human-written articles to a dedicated classifier, and evaluates it against LIFE alone and against a single model with no routing at all. Fourth, it shows that the routed system does not outperform the simplest baseline in this comparison — one BERT classifier trained on everything, provenance-blind — and traces that shortfall to some, measurable causes: the human branch of PRISM and the act of routing itself.

2. Background and Related Work

2.1 LLM-Generated Fake News Benchmarks

PolitiFact++ and GossipCop++ extend the PolitiFact and GossipCop portions of FakeNewsNet by pairing each human-written article with a GPT-3.5-generated counterpart, produced using what its authors call Structured Mimicry Prompting to keep the machine-generated text stylistically close to news rather than obviously synthetic (Su et al., 2024). PolitiFact++ contains 97 human-fake, 194 human-real, 97 machine-fake, and 132 machine-real articles (520 total); GossipCop++ contains 4,084 human-fake, 8,168 human-real, 4,084 machine-fake, and 4,169 machine-real articles (20,505 total). LIFE's own reported binary task, MF-vs-MR, uses only the two machine-generated subsets — 229 articles for PolitiFact++, 8,253 for GossipCop++ (Wang et al., 2026) — a smaller pool than the full four-subclass benchmark this paper works with from Chapter 4 onward.

The design of this benchmark already anticipates the central finding of this paper. Su et al. (2024) trained five BERT-family classifiers (BERT, RoBERTa, ELECTRA, ALBERT, DeBERTa) under five different mixtures of human- and machine-written fake news and found a consistent, asymmetric failure: a classifier trained exclusively on machine-generated fake news (100% MF) correctly identifies almost no human-written fake news — HF accuracy collapses to 12.12% on GossipCop++ and 9.28% on PolitiFact++ — even as the same classifier reaches near-ceiling accuracy on the machine-generated class it was trained on. The reverse case is far gentler: a classifier trained exclusively on human-written data still detects machine-generated fake news reasonably well. Su et al. (2024) attribute this to detectors keying on source-level cues — whether a passage carries the fingerprint of machine generation — rather than on the article's actual veracity. This paper reaches the same conclusion independently, using a completely different architecture (a perplexity-reconstruction fingerprint rather than a fine-tuned encoder) and a different diagnostic route. That two unrelated detection methods, tested on the same benchmark, fail in the same direction is the strongest evidence available that this asymmetry is a property of the task, not an artifact of either method.

2.2 LIFE: Prompt-Induced Linguistic Fingerprints

LIFE (Wang et al., 2026) detects LLM-generated fake news by analyzing the generation process itself rather than surface content. The method prepends a malicious prompt, instructing a reconstruction LLM to rewrite the following article as fake news, to each article, then records the model's per-token probability of reproducing the article's own original tokens under that prompt. Wang et al. (2026) find a consistent statistical divergence: original tokens from real news receive systematically lower reconstruction probability than original tokens from fake news, a pattern they term a linguistic fingerprint. A key-fragment extraction step selects the ten to fifteen

sentences whose removal most changes the model's veracity decision, and only those sentences' reconstruction probabilities are retained as a feature vector, classified by a CNN-Transformer sequence model with a sigmoid output trained under binary cross-entropy loss (Eq. 11–12 in the source paper). Wang et al. (2026) report target performance of Acc 0.900 / Macro-F1 0.882 on PolitiFact++ and Acc 0.937 / Macro-F1 0.924 on GossipCop++, both on the binary MF-vs-MR task, and show LIFE outperforming ten baselines including fine-tuned BERT, RoBERTa, and LLaMA2-7B on this task.

2.3 Machine-Generated Text Detection

Distinguishing AI-generated text from human-written text is, by itself, a largely separate and increasingly tractable problem from determining whether a claim is true. Masih et al. (2025) trained a battery of classical, recurrent, and transformer models to separate GPT-3.5/GPT-4 output from human writing on a balanced 20,000-sample corpus, finding that RoBERTa reached 96.1% accuracy, with calibration and explainability analysis (LIME, SHAP) showing the model relies on structural cues — connector words, sentence uniformity — rather than content. Raza et al. (2025) ran a closely related comparison specific to fake news: BERT-like encoder models (RoBERTa 89.68% F1) against decoder-only LLMs fine-tuned as classifiers (Llama2-7B, Mistral-7B, both under 81% F1), and found encoder-only classification consistently stronger, while the LLM-based classifiers were more robust to adversarial text perturbations. That an encoder-only classifier beats a generative-LLM-based approach at the classification task itself, even though the LLM brings other strengths, is the same shape of result this paper arrives at independently in Chapter 6, using a different method entirely.

2.4 Conventional Fake-vs-Real Classifiers

This paper's practical system, PRISM (Chapter 5), routes human-written articles to a plain BERT classifier, and its results (Chapter 6) compare that system against a single BERT model with no routing at all. Both choices are grounded in an established line of work applying BERT directly to fake news classification. Rai et al. (2022) combined BERT with an LSTM layer over news titles from the original FakeNewsNet PolitiFact and GossipCop subsets, reporting 88.75% accuracy on PolitiFact and 84.10% on GossipCop — a modest but consistent improvement over a vanilla fine-tuned BERT baseline (86.25% / 83.00%) and over older content- and social-context-based methods such as TCNN-URG, CSI, and HAN. Oad et al. (2024) extended BERT with additional linear, dropout, and activation layers and reported 85% accuracy on the BuzzFeed dataset and 98.23% on a self-collected regional Twitter dataset, again outperforming a vanilla BERT baseline. Both results confirm that even a relatively simple, provenance-blind BERT classifier is a strong baseline on human-written fake news — the exact population this paper's diagnostic chain shows LIFE cannot handle. Devlin et al. (2019) is the foundational reference for BERT itself, used throughout this paper's own PRISM and monolithic classifiers.

3. Reproducing LIFE

This paper implements a paper-faithful version of LIFE's classifier head — the sigmoid-and-binary-cross-entropy formulation given in Eq. 11–12 of Wang et al. (2026) — over the same CNN-Transformer trunk, rather than the released repository's sequence-tagging head, so that the same classifier design carries through every experiment in this paper, from reproduction to the final routed system. All experiments reported in this paper, including the diagnostic chain in Chapter 4 and the routed system, PRISM, in Chapters 5–6, were run on Google Colab using a single NVIDIA A100 GPU. On PolitiFact++'s binary MF-vs-MR task, following the paper's own

80/20 split convention, this reproduction reaches Accuracy $86.82\% \pm 2.16$, Macro-F1 $84.48\% \pm 2.99$ (mean $\pm$ standard deviation over 20 seeds for 50 epochs each), against Wang et al.'s (2026) reported target of Accuracy 90.0% / Macro-F1 88.2%. The gap, roughly 3.2 points of accuracy and 3.7 points of Macro-F1, is not exact reproduction, but it is close: numbers are not exactly the same, but the difference is slim. On GossipCop, reproduction reaches an accuracy of 93.4% with a Macro-F1 score of 93.2% (no seeding is necessary as dataset is large enough; data is trained for 15 epochs), against Wang et al.'s (2026) reported target of Accuracy 93.7% / Macro-F1 92.4%. The reproduction more or less matches the performance the original paper reports on the same task.

This result establishes the baseline this paper's diagnostic chain works from. LIFE, faithfully reimplemented, does what it claims to do on the task it was built for: separating text a large language model was instructed to rewrite as fake from text the same model was instructed to rewrite as real. What happens when the task changes — when the fake class includes articles written by humans — is the subject of the next chapter.

4. Why Build PRISM? A Diagnostic Chain

4.1 Combining Human and Machine Articles

The first departure from LIFE's original task is the most direct one: pool human-written and machine-generated articles into a single fake-vs-real classification, with fake defined by veracity (HF + MF) rather than by provenance (MF alone), and real defined the same way (HR + MR). This is the task most fake-news detectors are implicitly assumed to solve — real-world deployment cannot filter its input to LLM-generated text only. Training and testing LIFE's paper-faithful head on this combined population gives Accuracy $71.9\% \pm 1.2$, Macro-F1 $65.2\% \pm 2.1$

on PolitiFact++, and Accuracy 76.3%, Macro-F1 72.0% on GossipCop++. This shows that neither result is close to LIFE's binary MF-vs-MR performance from Chapter 3. Something is being lost when human-written fakes enter the fake class. The next section identifies exactly what.

4.2 Diagnosing Why: Four-Class Classification

To isolate the source of the drop, the four subclasses are kept separate: LIFE is trained to classify each article as human-fake, human-real, machine-fake, or machine-real, on both Politifact++ and GossipCop++. Overall performance is weak, Accuracy $46.6\% \pm 2.2$, Macro-F1 $39.7\% \pm 4.2$ on PolitiFact++, and Accuracy 60.7%, Macro-F1 49.1% on GossipCop++, but the aggregate numbers understate how uneven the failure is. Table 1 shows the precision and recall for each of the four classes.

Table 1: Four-class precision/recall (%)

Dataset	Human-Fake (HF)	Human-Real (HR)	Machine-Fake (MF)	Machine-Real (MR)
GossipCop++	0.0 / 0.0	53.5 / 90.5	85.0 / 90.9	57.9 / 32.0
Politifact++	0.0 / 0.0	43.9 / 78.4	55.6 / 88.2	45.5 / 14.7

The result is unambiguous: on GossipCop++, LIFE separates machine-fake from machine-real well (85.0% precision, 90.9% recall on MF) while being completely blind to human-written fake news, 0% precision, 0% recall. LIFE never once correctly labels a human-written fake article as fake. It does not systematically confuse HF with HR either; it simply never predicts the human-fake class in a way that lands on a true positive. This is precisely the failure Su et al. (2024)

documented across five different transformer architectures on the same benchmark family — the mechanism recurs even though the underlying detector here is not a fine-tuned encoder at all, but a perplexity-reconstruction fingerprint over an entirely different pipeline.

PolitiFact++'s pattern is noisier: alongside HF's 0/0, every other class sees a drop in precision and recall on this run. The most significant drop is the MF class, where it recorded 55.6% precision in comparison to GossipCop++'s 85.0%. This is an additional failure mode from the provenance-tracking one GossipCop++ illustrates cleanly: PolitiFact++'s four-class split leaves only around 78 training articles per class, and at that scale the model appears to partly collapse toward predicting the majority-adjacent class rather than learning any class-specific signal at all. GossipCop++, twenty times larger, is the clean illustration of the underlying mechanism; PolitiFact++ layers a small-sample collapse on top of it.

4.3 Confirming It: Human-Only Classification

The four-class result narrows the failure to one subclass, but it does not yet rule out that LIFE is simply struggling with a four-way decision boundary rather than failing on human-written fake news specifically. To isolate the claim, all machine-generated articles are removed entirely, leaving a binary human-fake-vs-human-real task (HF vs. HR) with no MF or MR examples anywhere in training or testing. This is a negative control: since LIFE's fingerprint is built by prompting a reconstruction LLM to rewrite an article as fake, and no LLM ever touched either class here, there should be no fingerprint for the model to find if the earlier diagnosis is correct.

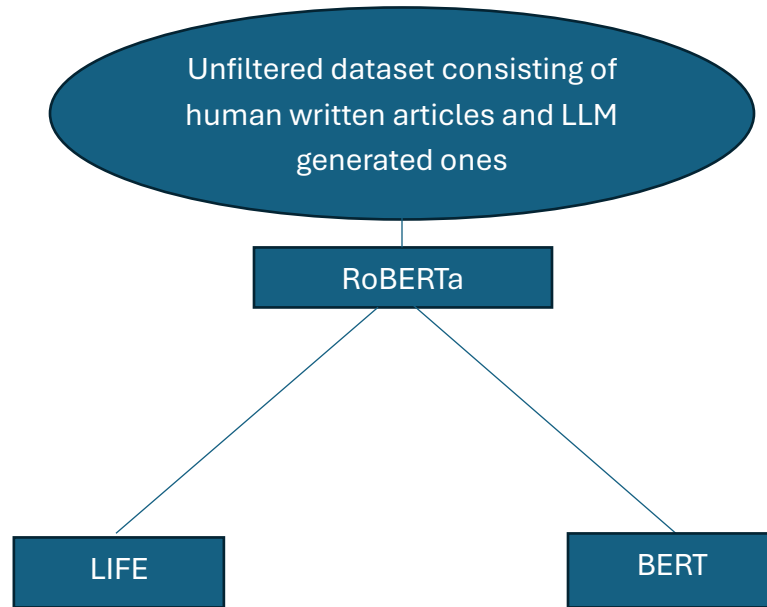
The result confirms it. Mean performance across a 20-seed sweep is Accuracy $72.17\% \pm 1.66$, Macro-F1 $45.18\% \pm 6.10$, and the individual-seed behavior is more informative than the mean: a representative collapsed seed reports Accuracy 1.2% / Macro-F1 41.6%, with real

precision/recall of 71.2% / 100.0% and fake precision/recall of exactly 0.0% / 0.0%, the model predicting "real" for every single article, regardless of ground truth. The wide standard deviation on Macro-F1 relative to the tight standard deviation on Accuracy is itself diagnostic: most seeds collapse fully to the prior, and only a few catch a handful of the human-written fakes in the test set, which moves Macro-F1 without moving Accuracy much. This is the same 0%/0% pattern the four-class diagnosis produced, now isolated from any possible confound with machine-generated text, on a task built specifically to have no fingerprint for LIFE to detect. LIFE cannot recognize human-written fake news — not in combination with machine-generated text, and not in isolation from it.

5. A Practical System: Provenance-Routed Detection

The diagnostic chain in Chapter 4 points to a specific design: rather than asking one detector to handle both machine-generated and human-written content, route each article to the detector suited to it. PRISM, Provenance-Routed Inference via Specialist Models, built here consists of three components. A RoBERTa-based provenance router is trained to classify each article as LLM-generated or human-written, using the same PolitiFact++/GossipCop++ articles with provenance as the label rather than veracity. Articles the router calls LLM-generated are passed to LIFE, trained and evaluated exactly as in Chapters 3–4 on its native task. Articles the router calls human-written are passed instead to a dedicated BERT classifier, fine-tuned directly on veracity within the human-written population. The two branches' outputs are combined into a single binary real-or-fake prediction for every article in the dataset, regardless of which branch produced it. This is the system-level accuracy that Chapter 6 reports and compares against alternatives.

PRISM illustration



6. Results

6.1 PRISM vs. LIFE Alone vs. BERT Alone

Table 2 reports whole-dataset, cross-validated GossipCop++ results (pooled across folds so that every one of the 20,505 articles is scored exactly once, by a model that never trained on it) for three configurations: PRISM described in Chapter 5; LIFE alone, applied to every article regardless of provenance; and a single BERT alone classifier trained on all four subclasses with no routing whatsoever.

Table 2: System comparison, GossipCop++, pooled cross-validated results

	GossipCop (Acc/F1)	Politifact (Acc/F1)
LIFE alone (MF+HF as “Fake” and MR+HR as “Real”)	76.3 / 72.0	71.9 / 65.2
PRISM	84.4 / 83.6	80.0 / 77.8
BERT Alone (MF+HF as “Fake” and MR+HR as “Real”)	85.16 / 84.7	85.0 / 83.6

Two comparisons follow from this table. First, PRISM clearly outperforms LIFE used alone on the whole dataset, a 9.3-point accuracy gain on GossipCop and 8.1-point accuracy gain on Politifact, which is exactly what the diagnostic chain predicts: LIFE's near-total blindness to human-written fake news, which makes up a large share of any realistic dataset, drags its whole-dataset accuracy down, and routing human-written articles away from it recovers most of that loss. Second, PRISM does not outperform the single BERT classifier, it trails by 0.8 accuracy points and 1.1 Macro-F1 points on GossipCop, and 5.0 accuracy points and 5.8 Macro-F1 points on Politifact. PRISM, explicitly engineered to route each article to a specialized detector, performs no better, on this measure, than a single provenance-blind classifier trained on everything. The next section identifies why.

6.2 Why: Component-Level Diagnosis

PRISM's two branches can be compared directly on the population each actually receives in deployment — the router's predicted-human pile for the BERT branch, and the router's predicted-LLM-generated pile for LIFE. On this apples-to-apples basis, the human-written BERT branch reaches Accuracy 80.78%, Macro-F1 77.92%, while LIFE's branch reaches Accuracy 89.44%, Macro-F1 89.42%. This shows LIFE outperforms BERT by 8.7 accuracy points and 11.5 Macro-F1 points. The human branch is the weaker of PRISM's two components, and by a wider margin on Macro-F1 than on raw accuracy, meaning the human branch's errors are not evenly spread but concentrated in ways that hurt the minority class harder than accuracy alone reveals.

This finding is dataset-dependent, and the dependency is itself informative. On PolitiFact++, this comparison reverses, the human branch (80.56% accuracy) actually beats LIFE's branch (77.71% accuracy). However, PolitiFact++'s smaller sample size (520 articles total) produces far higher seed-to-seed variance on LIFE's branch specifically (± 17.05 percentage points, against GossipCop++'s ± 1.83), so this reversal reflects instability rather than a genuine architecture difference. The honest reading is that the human-branch-drags-PRISM-down theory holds on the dataset large enough to trust the measurement, and is inconclusive on the smaller one.

A second, independent contributor to the PRISM's shortfall is misrouting. The router reaches 95.01% accuracy on the provenance classification task itself is high, but not perfect. Its confusion matrix over the pooled test set shows 711 of 12,252 true human-written articles (5.8%) misclassified as LLM-generated and sent to LIFE instead of to the human branch. Given LIFE's near-total inability to detect human-written fake news (Chapter 4.3), every one of those 711 misrouted articles that happens to be a human-written fake is very likely to be scored wrong

regardless of which branch nominally "should" have received it — the router's own error rate compounds directly into the weakest point of the system it feeds.

7. Discussion

7.1 Why LIFE Fails on Human Fake News

The diagnostic chain in Chapter 4 converges on a single explanation, and Chapter 6's component-level results confirm it holds even inside a deployed system. LIFE's detection signal is built entirely from a reconstruction fingerprint: how differently a large language model, prompted to rewrite an article as fake, treats the article's own original tokens. That signal exists only where an LLM's generation process is actually present in the text's history. Human-written fake news was never touched by an LLM at any stage, so there is no fingerprint there to find. This is not a training-data problem that more human-fake examples would fix: Chapter 4.1 trained LIFE directly on human-written fakes, labeled as fake, mixed with machine-generated fakes, and it still failed to learn the class. Su et al. (2024) documented the same pattern in an entirely different architecture family, which suggests the underlying limitation is inherent to any detector whose signal is anchored to a machine-generation process, LIFE included, rather than a shortcoming specific to this implementation.

The practical consequence follows directly. LIFE is best understood not as a fake-news detector in the general sense, but as a machine-generated-text detector that happens to correlate with fakeness on the specific, narrow task it was built for MF vs MR, where every article in both classes was written by the same kind of process and only the prompt differed. Route it correctly, and it performs exactly as advertised, as Chapter 6.2's 89.44% accuracy on its deployed pile shows.

7.2 Limitations

PolitiFact++'s small sample size is a recurring limitation throughout this paper, not a single localized weakness. The four-class diagnosis in Chapter 4.2 showed a distinguishable small-sample collapse pattern on PolitiFact++ that GossipCop++ did not exhibit, and the component-level comparison in Chapter 6.2 showed LIFE's own branch accuracy varying by roughly ten times more, seed to seed, on PolitiFact++ than on GossipCop++. Wherever PolitiFact++ and GossipCop++ results diverge in this paper, GossipCop++, twenty times larger, is the more trustworthy read.

8. Conclusion

This paper set out to test whether a fake-news detector built around large-language-model generation dynamics could do the job its name promises: detect fake news. The reproduction in Chapter 3 confirms LIFE does what it was designed to do, closely matching its reported performance on the machine-generated-only task it was built for. Every step past that point, combining human and machine articles, decomposing the result by class, and finally isolating human-only data completely, shows the same failure recurring in the same place: LIFE cannot detect human-written fake news, under any training configuration tested. Building PRISM, a provenance-routed system around this finding recovers most of the performance LIFE alone loses by being applied indiscriminately, but does not beat the simplest available alternative, a single BERT classifier with no routing at all, because the human branch of PRISM is itself the weaker of its two halves, and because a small but real share of human-written articles are misrouted into the one branch guaranteed to miss them. The result across all of this is not that provenance-based detection is worthless. LIFE separates machine-generated fake news from machine-generated real news cleanly, and a router can identify which articles it should be trusted

on with 95% accuracy. The result is narrower and more specific: a detector that keys on how text was generated should not be asked, and should not be marketed, as a detector of whether text is true.

Several directions remain unexplored. This paper did not measure computational cost: LIFE's reconstruction step requires a full forward pass through a 7-billion-parameter language model per article, architecturally far more expensive than the BERT forward. It also did not test PRISM as a multiclass system — Chapter 4.2 uses the four-class task (HF/HR/MF/MR) only as a diagnostic on LIFE in isolation, not as PRISM's deployed output, which currently collapses to a binary real/fake decision regardless of provenance; whether a provenance-aware four-way output changes the practical picture is untested. Third, this paper evaluated only one human-branch classifier, a plain fine-tuned BERT, against LIFE; whether a stronger human-fake detector, such as the BERT+LSTM architecture of Rai et al. (2022) or the layer-augmented Enhanced BERT of Oad et al. (2024), would close the gap between PRISM and the single BERT baseline in Chapter 6.1 is a direct, testable extension of the finding that the human branch is the weaker half. Finally, this paper's scope is limited to text; extending either LIFE or PRISM to multimodal fake news, video, audio, or image-paired content, is a substantially different problem, since LIFE's core signal depends on token-level language-model reconstruction probabilities that have no direct analogue in other modalities, and remains open.

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